



NORTH SYDNEY BOYS HIGH SCHOOL

2010 Year 11 Task 2

Mathematics

Extension 1

General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Weiss

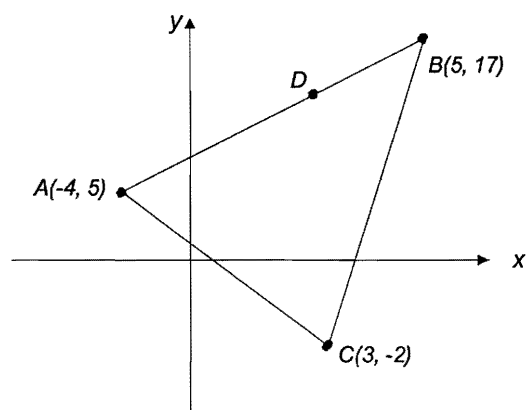
Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total %
Mark	$\overline{10}$	$\overline{9}$	$\overline{17}$	$\overline{10}$	$\overline{8}$	$\overline{54}$	$\overline{100}$

Question 1 (10 marks)

Marks



The triangle ABC has vertices $A(-4, 5)$, $B(5, 17)$ and $C(3, -2)$.

- (a) Show that the equation of the straight line through A and C is $x + y - 1 = 0$ 2
- (b) Find the length of side AC . 1
- (c) Point D divides side AB internally in the ratio $2 : 1$.
Find the co-ordinates of D . 2
- (d) Find the perpendicular distance of B from AC . 2
- (e) Hence calculate the area of triangle ΔABC 2
- (f) The line through D parallel to AC meets BC at point E .
Calculate the area of triangle ΔBDE . 1

Question 2 (9 marks)

- (a) Find the equation of the line which passes through the point of intersection of $15x - 6y - 10 = 0$ and $x + y - 3 = 0$, and through the point $P(1, 3)$. 3
- (b) Find the angle that the line $2x + 3y + 1 = 0$ makes with the positive direction of the x -axis. 2
- (c) Sketch the region where $y \leq x^2 - 4$ and $y > -3x$ are true simultaneously. Show any boundary points clearly. 4

Question 3 (17 marks)

(a) Express in simplest form as a single fraction, with no negative indices:

(i) $4x^{-2}$	(ii) $\frac{x^{-2}-1}{x^{-1}-1}$	3
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(b) Solve for x :

(i) $9^x = 27$	(ii) $x^{\frac{-5}{4}} = 32$	4
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(c) Evaluate:

(i) $\log_a \frac{1}{\sqrt{a}}$	(ii) $\log_3 63 - \log_3 7$	3
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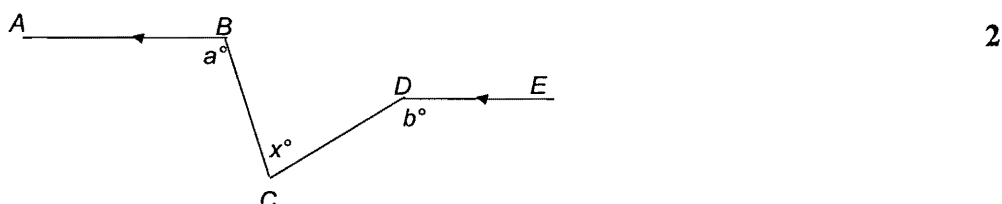
(d) Evaluate $\log_4 27$ to two decimal places.	2
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(e) Solve for x :

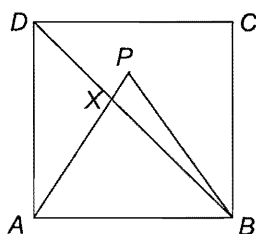
(i) $\log_{11} 2 + \log_{11} x = \log_{11} 7$	(ii) $2 \log_5 x = \log_5 (4x + 5)$	5
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Question 4 (10 marks)

(a) Express x in terms of a and b . (*no reasons needed*)



(b)



$ABCD$ is a square.

PAB is an equilateral triangle.

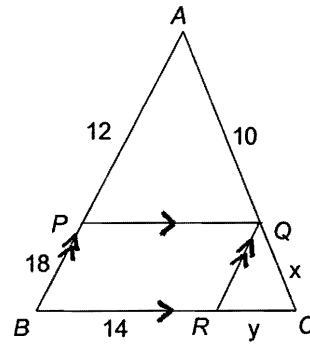
Calculate the size of $\angle DXA$, giving geometrical reasons.

(c) Find the size of each interior angle in a regular 20-sided polygon.	2
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- (d) Find the values of x and y .
Give the geometrical reason.

3

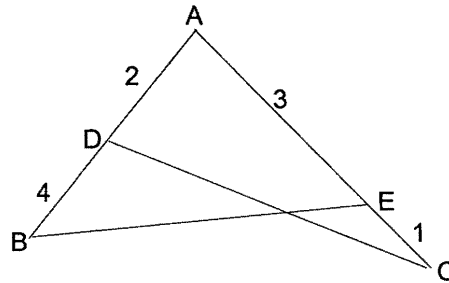
(Not to scale)



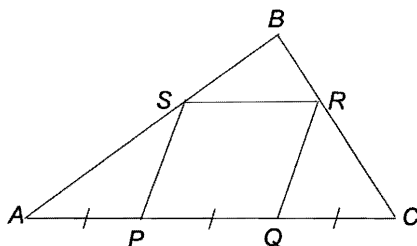
Question 5 (8 marks)

- (a) (i) State the geometrical reason why $\triangle ABE$ is similar to $\triangle ACD$ in the diagram below. 1
(ii) Given that $BE = 5$, calculate DC . 2

(Not to scale)



- (b) In the following diagram $PQRS$ is a rhombus.



- (i) Prove $\angle SPQ = 2 \times \angle SAP$ 2
(ii) Prove $\angle ABC = 90^\circ$ 3

END OF EXAMINATION

Suggested Solutions

Question 1

(d) (2 marks)

(a) (2 marks)

- ✓ [1] for correct gradient $m = -1$.
- ✓ [1] for setting up equation.

- ✓ [1] for correct substitution into $\perp$ dist. formula.
- ✓ [1] for final answer.

$$\begin{aligned}
 A(-4, 5) \quad C(3, -2) \\
 \frac{y+2}{x-3} &= \frac{5+2}{-4-3} \\
 &= \frac{7}{-7} = -1 \\
 \therefore y+2 &= -x+3 \\
 y &= -x+1 \\
 x+y-1 &= 0
 \end{aligned}$$

$$\begin{aligned}
 AC : x+y-1=0 \quad B(5, 17) \\
 d_{\perp} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\
 &= \frac{|1(5) + 1(17) - 1|}{\sqrt{1^2 + 1^2}} \\
 &= \frac{21}{\sqrt{2}} = \frac{21\sqrt{2}}{2}
 \end{aligned}$$

Alternatively, substitute coordinates of points A and C into equation.

(e) (2 marks)

(b) (1 mark)

- ✓ [1] for correct substitution.
- ✓ [1] for final answer.

$$\begin{aligned}
 d_{AC} &= \sqrt{(5 - (-2))^2 + (-4 - 3)^2} \\
 &= \sqrt{7^2 + 7^2} \\
 &= \sqrt{49 \times 2} = 7\sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 A &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times 7\sqrt{2} \times \frac{21}{\sqrt{2}} \\
 &= 73.5
 \end{aligned}$$

(c) (2 marks)

- ✓ [1] for each value of x and y correctly found.

$$\begin{array}{ccc}
 A(-4, 5) & & B(5, 17) \\
 & \swarrow \quad \searrow & \\
 2 & & -1
 \end{array}$$

(f) (1 mark)

$$\begin{aligned}
 x_D &= \frac{1(-4) + 2(5)}{2+1} \quad \left| \quad y_D = \frac{1(5) + 2(17)}{2+1} \right. \\
 &= \frac{-4+10}{3} = 2 \quad \left| \quad = \frac{5+34}{3} = 13 \right. \\
 &\therefore D(2, 13)
 \end{aligned}$$

$$\begin{aligned}
 \frac{A_{\triangle BDE}}{A_{\triangle ABC}} &= \frac{1^2}{3^2} = \frac{1}{9} \\
 \therefore A_{\triangle BDE} &= \frac{1}{9} \times \frac{147}{2} = \frac{147}{18} \\
 &= \frac{49}{6} = 8\frac{1}{6}
 \end{aligned}$$

Other methods exist, though this one shown here would be the easiest.

Question 2

(a) (3 marks)

- ✓ [1] for correctly placing equations into k method equation.
- ✓ [1] for $k = 13$.
- ✓ [1] for final factorised equation.

$$\begin{aligned}
 15x - 6y - 10 &= 0 & x + y - 3 &= 0 \\
 15x - 6y - 10 + k(x + y - 3) &= 0 \Big|_{x=1, y=3} \\
 15 - 18 - 10 + k(1 + 3 - 3) &= 0 \\
 k &= 18 + 10 - 15 \\
 &= 13 \\
 15x - 6y - 10 + 13x + 13y - 39 &= 0 \\
 28x + 7y - 49 &= 0 \\
 7(4x + y - 7) &= 0 \\
 \therefore 4x + y - 7 &= 0
 \end{aligned}$$

(b) (2 marks)

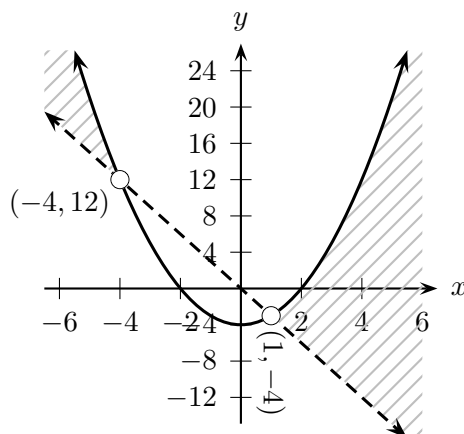
- ✓ [1] for $\tan \theta = -\frac{2}{3}$.
- ✓ [1] for final answer.

$$\begin{aligned}
 2x + 3y + 1 &= 0 \\
 3y &= -2x - 1 \\
 y &= -\frac{2}{3}x - \frac{1}{3} \\
 \therefore m &= -\frac{2}{3} = \tan \theta \\
 \theta &= 180^\circ - 33^\circ 41' \\
 &= 146.31^\circ / 146^\circ 19'
 \end{aligned}$$

(c) (4 marks)

- ✓ [1] for dotted line $y = -3x$
- ✓ [1] for parabola $y = x^2 - 4$.
- ✓ [1] for boundary points
- ✓ [1] for correct regions.

$$\begin{aligned}
 \begin{cases} y = x^2 - 4 \\ y = -3x \end{cases} \\
 x^2 - 4 &= -3x \\
 x^2 + 3x - 4 &= 0 \\
 (x + 4)(x - 1) &= 0 \\
 \therefore x &= -4, 1
 \end{aligned}$$

**Question 3**

(a) i. (1 mark)

$$4x^{-2} = \frac{4}{x^2}$$

ii. (2 marks)

- ✓ [1] for correctly multiplying numerator & denominator by x^2 to obtain $\frac{1-x^2}{x-x^2}$.
- ✓ [1] for final answer.

$$\begin{aligned}
 \frac{x^{-2} - 1}{x^{-1} - 1} &= \frac{\frac{1}{x^2} - 1}{\frac{1}{x} - 1} = \frac{1 - x^2}{x - x^2} \\
 &= \frac{(1-x)(1+x)}{x(1-x)} \\
 &= \frac{1+x}{x}
 \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for $3^{2x} = 3^3$.
- ✓ [1] for final answer.

$$9^x = 27$$

$$3^{2x} = 3^3$$

$$2x = 3$$

$$x = \frac{3}{2}$$

ii. (2 marks)

- ✓ [1] for $x = 32^{-4/5}$.
- ✓ [1] for final answer.

$$x^{-\frac{5}{4}} = 32$$

$$\left(x^{-\frac{5}{4}}\right)^{-\frac{4}{5}} = (32)^{-\frac{4}{5}}$$

$$x = 2^{-4} = \frac{1}{16}$$

(c) i. (1 mark)

$$\log_a \frac{1}{\sqrt{a}} = \log_a a^{-\frac{1}{2}} = -\frac{1}{2}$$

ii. (2 marks)

$$\begin{aligned} \log_3 63 - \log_3 7 &= \log_3 \left(\frac{63}{7} \right) \\ &= \log_3 9 = 2 \end{aligned}$$

(d) (2 marks)

✓ [1] for correct change of base.

✓ [1] for final answer.

$$\begin{aligned} \log_4 27 &= \frac{\log_{10} 27}{\log_{10} 4} \\ &= 2.38 \text{ (2 d.p.)} \end{aligned}$$

(e) i. (2 marks)

✓ [1] for correct use of addition rule for logarithms.

✓ [1] for final answer.

$$\begin{aligned} \log_{11} 2 + \log_{11} x &= \log_{11} 7 \\ \log_{11} (2x) &= \log_{11} 7 \\ \therefore 2x &= 7 \\ x &= \frac{7}{2} \end{aligned}$$

ii. (3 marks)

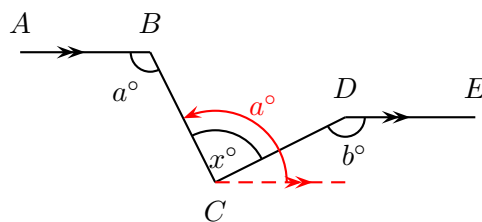
✓ [1] for $x^2 = 4x + 5$.✓ [1] for solutions of x .✓ [1] for justification of $x = 3$.

$$\begin{aligned} 2 \log_5 x &= \log_5 (4x + 5) \\ \log_5 (x^2) &= \log_5 (4x + 5) \\ x^2 &= 4x + 5 \\ x^2 - 4x - 5 &= 0 \\ (x - 5)(x + 1) &= 0 \\ x &= 5, -1 \end{aligned}$$

But as $x > 0$ in the domain of the logarithmic function. Hence $x = 5$.

Question 4

(a) (2 marks)



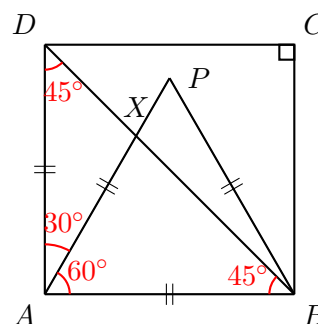
$$a = x + (180^\circ - b)$$

$$x = a + b - 180^\circ$$

(b) (3 marks)

✓ [2] for correct reasoning.

✓ [1] for final answer.



- $\angle PAB = 60^\circ$ ($\angle$ in equilateral $\triangle$)
- $\therefore \angle XAD = 30^\circ$.
- $\angle ADB = 45^\circ$
(diagonals of a square bisect the $\angle$)
- By the $\angle$ sum of $\triangle DXA$,

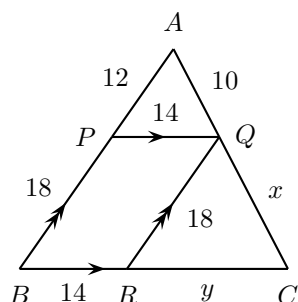
$$\angle DXA = 180^\circ - 30^\circ - 45^\circ = 105^\circ$$

(c) (2 marks)

$$\begin{aligned} \sum \angle &= (n - 2) \times 180^\circ \\ &= 18 \times 180^\circ \\ &= 3\,240^\circ \\ \therefore \angle &= \frac{3\,240^\circ}{20} = 162^\circ \end{aligned}$$

(d) (3 marks)

- ✓ [1] each for correct x and y .
- ✓ [1] for correct reasoning.



- By the intercepts of transversals over (b) parallel lines PQ and BC ,

$$\frac{12}{18} = \frac{10}{x} \Rightarrow x = \frac{10 \times 3}{2} = 15$$

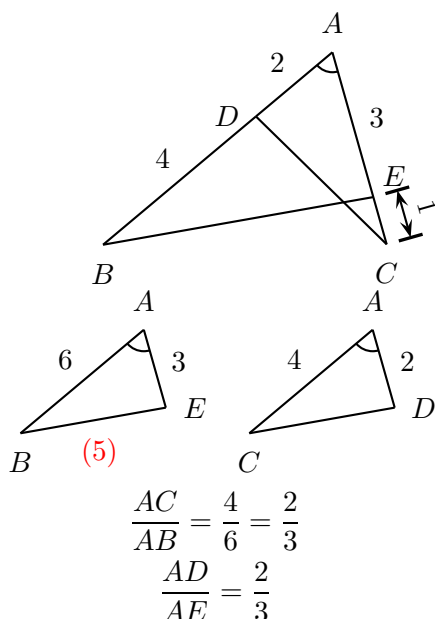
- By the same reason as above,

$$\frac{y}{14} = \frac{x}{10} \Rightarrow y = \frac{15 \times 14}{10} = \frac{3 \times 3 \times 7}{2} = 21$$

Question 5

(a) i. (1 mark)

- ✓ [1] for final statement. Proof shown for completeness.
- Splitting the triangles into separate shapes:



- $\angle BAE = \angle DAC$ (common from diagram)

Hence $\triangle ABE \parallel \triangle ACD$ (two matching sides in same ratio plus included angle equal)*.

ii. (2 marks)

$$\begin{aligned} BE &= 5 \\ \frac{CD}{BE} &= \frac{2}{3} \Rightarrow \frac{CD}{5} = \frac{2}{3} \\ CD &= \frac{10}{3} \end{aligned}$$

i. (2 marks)

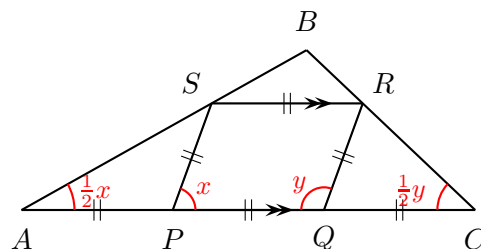
- ✓ [2] for correct proof shown.
- Let $\angle SPQ = \alpha$.
- Since $\angle PAS = \angle PSA$ (base $\angle$ of isos $\triangle$), and $\angle PAS + \angle PSA = \angle SPQ$ (exterior $\angle$ of $\triangle$)

$$2\angle PAS = \angle SPQ$$

$$\therefore \angle PAS = \frac{1}{2}\angle SPQ = \frac{1}{2}x$$

ii. (3 marks)

- ✓ [1] for each matching bullet point shown.



- By the same reasoning as the previous part, if $\angle RQP = y$ then $\angle RCQ = \frac{1}{2}y$.
- Since $\angle SPQ + \angle RQP = 180^\circ$ (cointerior $\angle$, $SR \parallel PQ$), then

$$\begin{aligned} \frac{1}{2}x + \frac{1}{2}y &= \frac{1}{2}(x + y) \\ &= 90^\circ \end{aligned}$$

- By the $\angle$ sum of $\triangle ABC$, if $\angle BAC + \angle BCA = 90^\circ$, then

$$\angle ABC = 90^\circ$$

*"SAS" will not be accepted for this part